

KOTA, India

WMO: 424520

Lat: 25.160N

Lon: 75.846E

Elev: 896

StdP: 14.23

Time Zone: 5.50 (IND)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.7	51.8	20.8	16.1	91.9	25.8	20.5	90.6	9.3	76.0	7.3	70.5	0.8	90	0.344

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	20.0	111.4	69.5	109.1	69.3	106.9	69.1	81.6	88.5	80.9	87.9	80.1	87.2	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.0	161.1	85.1	79.2	156.5	84.2	78.5	153.1	83.7	46.5	88.3	45.6	87.9	44.8	87.3	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.1	9.1	45.5	114.6	3.1	2.5	43.3	116.4	41.5	117.8	39.8	119.2	37.5	121.0		
			41.4	84.1	2.4	2.0	39.7	85.6	38.3	86.8	37.0	87.9	35.2	89.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.6	62.9	70.0	80.7	91.5	97.8	94.7	86.6	83.8	85.2	83.9	75.0	66.3
	DBStd	11.48	4.57	5.46	6.05	4.51	3.95	5.59	4.20	2.64	3.16	3.58	4.07	4.65
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	152	98	14	0	0	0	0	0	0	0	0	0	40
	CDD50	11523	400	561	952	1245	1481	1340	1135	1048	1057	1050	750	504
	CDD65	6202	33	155	487	795	1016	890	670	583	607	585	301	80
	CDH74	86433	220	1248	5879	12708	17736	14650	9040	6962	7779	7164	2532	515
	CDH80	51616	25	383	3112	8590	13285	10364	4690	2735	3737	3713	905	77
(14) Wind	WSAvg	2.9	1.6	2.0	2.4	3.2	4.8	5.5	4.8	3.8	3.0	1.6	1.1	1.2
(15) Precipitation	PrecAvg	27.6	0.2	0.2	0.1	0.1	0.4	2.9	10.0	9.2	3.5	0.6	0.0	0.2
	PrecMax	55.7	1.6	1.1	1.3	0.8	2.5	7.9	26.6	18.2	14.5	4.6	0.8	1.7
	PrecMin	16.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3	0.9	0.0	0.0	0.0	0.0
	PrecStd	10.9	0.4	0.3	0.3	0.2	0.5	2.3	5.8	4.4	3.0	1.1	0.2	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.0	91.7	103.8	110.5	114.2	114.1	105.2	95.9	99.8	100.0	92.7	85.4
		MCWB	62.0	63.2	64.1	66.3	69.3	71.2	75.7	77.4	72.8	67.6	63.6	63.1
	2%	DB	79.1	88.2	99.8	107.6	112.1	110.9	99.9	93.3	97.0	97.8	90.0	82.0
		MCWB	61.1	62.9	63.2	65.9	69.2	71.1	76.7	78.0	74.1	68.3	64.0	61.5
	5%	DB	76.2	85.0	96.7	105.5	110.3	107.8	96.6	91.2	95.1	95.7	87.9	79.5
		MCWB	59.7	61.7	63.3	65.7	69.1	72.0	77.2	78.4	74.4	68.2	63.5	60.4
	10%	DB	73.1	81.6	93.7	103.3	108.3	104.9	94.0	89.5	93.0	93.7	85.6	76.7
		MCWB	58.0	60.8	63.0	65.2	68.7	72.9	77.7	78.7	75.0	67.6	62.8	59.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	68.0	69.7	71.4	77.8	82.4	82.5	82.4	82.7	78.7	74.1	66.5
		MCDB	77.9	80.9	86.9	100.2	97.0	92.7	89.2	88.5	89.1	87.3	80.8	78.4
	2%	WB	63.5	66.0	67.9	69.7	76.0	80.0	81.6	81.4	80.7	76.9	70.5	64.4
		MCDB	74.8	81.0	88.1	97.7	97.7	91.6	88.6	87.4	87.3	84.4	77.9	76.9
	5%	WB	61.4	64.3	66.4	68.5	74.5	79.1	80.9	80.7	79.6	75.2	67.3	62.5
		MCDB	71.7	79.2	87.3	96.5	98.5	91.2	88.1	86.6	86.3	84.1	80.1	75.3
	10%	WB	59.7	62.7	65.0	67.4	73.2	78.4	80.2	80.0	78.7	73.2	65.3	60.9
		MCDB	69.8	77.7	87.0	95.8	98.8	91.0	87.4	85.7	85.6	85.3	80.7	73.7
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.6	21.4	22.0	20.0	16.6	11.0	9.5	12.9	19.7	20.6	19.7
		MCDBR	22.0	22.6	24.1	23.5	21.0	18.7	15.1	12.7	16.3	21.2	22.1	22.2
		MCWBR	9.5	8.4	7.5	6.9	6.2	5.8	3.7	3.8	4.8	5.9	6.7	8.0
	5% WB	MCDBR	19.3	19.8	20.9	21.6	19.7	15.7	11.3	10.2	11.8	14.8	17.2	19.2
		MCWBR	9.7	9.1	8.3	7.2	6.8	5.0	4.5	4.2	5.0	5.5	6.9	8.0
(40) Clear-Sky Solar Irradiance	taub		0.443	0.434	0.470	0.536	0.591	0.666	0.674	0.608	0.539	0.526	0.531	0.487
	taud		2.011	2.085	2.004	1.879	1.802	1.700	1.708	1.863	1.998	1.918	1.844	1.906
	Ebn at Noon		251	266	263	248	234	215	214	228	241	235	221	229
	Edh at Noon		49	49	56	65	70	77	76	65	56	57	58	53
(44) All-Sky Solar Radiation	RadAvg		1293	1622	1983	2233	2273	1959	1448	1397	1676	1687	1377	1239
	RadStd		100	65	70	63	59	134	174	141	145	80	95	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page